

Pathmika Weeraratna

Associate Tech Lead | Full-Stack Engineer (Mobile & Web)

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LinkedIn | GitHub

Summary

Full-stack engineering leader with nearly five years of experience shipping production mobile and web applications for telecom operators, having grown through every engineering level at Omobio into Associate Tech Lead. Contributed to 25+ client and product engagements – delivery, architecture, and pre-sales solutioning – across 9+ countries in Asia, Africa, and Latin America. Deep hands-on expertise spanning React Native mobile development (native modules, app security, monetization), Next.js/React web platforms, and REST API/backend design, alongside a track record of mentoring teams and owning delivery from architecture through production. Deepening technical foundations through an MSc in Artificial Intelligence at the University of Moratuwa. Recognized as Emerging Employee of the Year (2023).

Experience

Associate Tech Lead

Jan 2026 – Present

Omobio (Pvt) Ltd

- Lead, mentor, and scale a cross-functional engineering team of 10+ developers, directing solution design and delivery across multiple product initiatives.
- Serve as technical subject-matter expert for pre-sales engagements, producing architecture proposals and technical validation that support client acquisition.
- Lead two flagship full-stack platforms: the SMS Firewall AI/ML Suite (backend ML services integrated with reporting dashboards) and the SOR (Steering of Roaming) centralized reporting platform.
- Leverage AI tools throughout the design and development workflow to boost personal productivity, enabling faster turnaround and higher-quality output.
- Manage client engagements end-to-end, from requirements gathering through production delivery.

Senior Software Engineer

Jan 2024 – Jan 2026

Omobio (Pvt) Ltd

- Led Agile delivery of end-to-end mobile and web applications, owning architecture design, effort estimation, and SOW reviews for on-time, client-aligned outcomes.
- Delivered SQL/CDR reporting engineering for Banglalink's SMSC billing platform (Bangladesh), recovering ~85,000 dropped records and fixing a 240x query-performance regression (4+ min → ~20 sec).
- Architected cross-platform mobile solutions in React Native with deep native integration (Java, Kotlin, Swift), including a ~2,700-line custom native alarm engine built from the ground up across 26 files, replacing third-party alarm libraries on Android and iOS.
- Integrated RevenueCat for end-to-end in-app subscription purchasing across iOS StoreKit and Google Play Billing.
- Built HOPP, Hutch's Next.js 14 web self-care portal: authoring the core auth/locale middleware, a custom Axios API layer, and end-to-end payment-gateway integration across six payment methods.
- Continued ownership of Hutch Selfcare (led for 2+ years across this role and the one prior): led its React Native version upgrade from 0.66.5 to 0.76.5 and delivered 3+ change requests for new feature additions.
- Mentored a team of 10 junior developers, conducted code reviews, and established coding standards and GitHub Actions CI/CD pipelines to improve software reliability.

Software Engineer

Apr 2022 – Jan 2024

Omobio (Pvt) Ltd

- Led design and development of cross-platform mobile applications (Hutch Selfcare, Vodafone Selfcare, MyDialog, RetailHub, Dtel, Circadian, SFA) in React Native, managing releases across Play Store, App Store, and Huawei AppGallery.
- Hardened Hutch Selfcare against fraud with certificate/public-key pinning and a device-integrity gate, and built a dual FCM/HMS push-notification pipeline for Android.
- Led the React Native version upgrade of Vodafone Fiji's production self-care app from RN 0.59.3 to 0.70.5 with New Architecture scaffolding.
- Participated in pre-sales engagement and contributed to ISO 27001 readiness documentation and a company-wide Git branching standard.

- Received the Emerging Employee of the Year Award (2023) for outstanding technical contribution.

Software Engineer (Intern) – Full Stack

Oct 2021 – Apr 2022

Omobio (Pvt) Ltd

- Supported frontend, backend, and mobile feature development for enterprise-grade projects using React.js, React Native, Python, and RESTful APIs; transitioned to a full-time Software Engineer role based on strong performance.

Key Projects

- **Hutch Selfcare (Telecom Self-Care App), 2+ years:** Owned and evolved Hutch's flagship React Native app: security hardening, dual FCM/HMS push architecture, a RN version upgrade (0.66.5 → 0.76.5), and 3+ feature change requests.
- **HOPP Web Self-Care Portal, 2024:** Next.js/TypeScript portal with custom Axios API layer and payment-gateway integration, delivered under a Hybrid Git Flow model.
- **SMS Firewall AI/ML Suite:** Lead a modular AI/ML detection suite (FastAPI, containerized modules, single admin CMS) for OTP-bypass, phishing-URL, encoded-Unicode, grey-route, and campaign detection; an XGBoost/CatBoost ensemble powers phishing-URL detection, the OTP-bypass/grey-route and encoded-Unicode modules layer in semantic embeddings, and a local-LLM GenAI layer supports content review and translation across the suite.
- **SOR (Centralized Reporting Platform):** Lead a centralized telecom data-reporting platform spanning backend pipelines and web dashboards, with asynchronous task processing and containerized deployment.
- **FirewallBI (Telecom Reporting Platform), 2023–2025:** Multi-tenant BI platform deployed across 8 on-prem operator sites (live references at 1,500 TPS/16.3M subscribers and 500 TPS/3.5M subscribers), containerized with Docker and GitLab CI/CD.
- **MDA (Dialog Device Management Platform):** PHP Yii2 backend with a React/Redux-Saga admin console for carrier-grade IMEI blacklist/whitelist (EIR) device management, integrated with an Erlang/OTP telecom-signaling core.
- **Robi Bangla SMS Opt-In/Opt-Out Platform, 2023:** Reviewed the SRS and contributed solution architecture for a BTRC-mandated language-preference platform (Bangladesh): a React/PHP Yii2/Erlang stack serving 5 user roles with sub-100ms REST APIs.

Additional Experience & Engagements

Beyond full-stack mobile and web delivery, self-initiated the AI/ML fraud-detection work described above from the ground up. Also contributed to 13+ further engagements, including Dialog's CMDP, RetailHub, WinBack, and DF platforms (Sri Lanka); the RoamerSteering ELK analytics platform (2023); and solution proposals for Ncell (Nepal), Grameenphone (Bangladesh), Ethio Telecom (Ethiopia), and BSNL (India).

Areas of Expertise

- **Mobile:** React Native (incl. New Architecture), Android (Java/Kotlin), iOS (Swift/Objective-C), Huawei HMS, native module bridging
- **Web:** React.js, Next.js, TypeScript/JavaScript, Redux/Redux-Saga
- **Backend & APIs:** Python, FastAPI, REST API design, Microservices, Java (Spring Boot), .NET (ASP.NET Core)
- **App & Web Security:** SSL/certificate pinning, device-integrity checks, OAuth2, payment-gateway integration
- **Data Engineering:** PostgreSQL, ClickHouse, MariaDB/MySQL, Elasticsearch/ELK, Apache Airflow, Celery
- **DevOps:** Docker, Git, CI/CD (GitHub Actions, GitLab CI, Jenkins), Linux, Prometheus, Grafana
- **Leadership:** Agile delivery, technical mentoring, pre-sales & client engagement, architecture ownership

Education

Master of Science in Artificial Intelligence (Reading) – University of Moratuwa

July 2025 – Present (Expected 2027)

B.Sc. in Information Systems, Second Class Upper Division, GPA 3.44 – University of Colombo School of Computing
2019 – 2022

Diploma in English (Certificate Level), Distinction Pass – Seagis Campus, Nugegoda

2018

Achievements

- Emerging Employee of the Year – Omobio (2023): recognized for exceptional job performance and contribution to key software engineering projects.
- Director's List Awardee – University of Colombo (2021): achieved GPA above 3.7 in first academic year.

References available upon request.